

VDMMBESP User's Guide

Version 1.0

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1. Introduction

VDMMBESP is a batch-simulation tool developed with MATLAB App Designer. Integrated with the COMSOL Multiphysics finite element simulation engine, this software is designed to simulate and analysis the potential distribution characteristics of metal pipelines with pitting defects under electric field excitation. Combined with data analysis and machine learning capabilities, it forms a complete workflow that covers physical modelling, batch sample generation, statistical analysis, and predictive modelling.

2. Installation and Startup

2.1 Environment Requirements

The hardware and software environment requirements for running the program are specified in the table below:

Item	Requirement	Remark
Operating System	Windows 10/11 (64-bit)	Windows 11 is recommended
MATLAB	R2022a or later	App Designer component is required
COMSOL Multiphysics	COMSOL Multiphysics 6.2 with LiveLink for MATLAB	For finite element electric field simulation calculation
Database (Optional)	MySQL 8.0+ and ODBC Driver	For long-term storage and multi-user shared querying of simulation results

2.2 Launch MATLAB via COMSOL

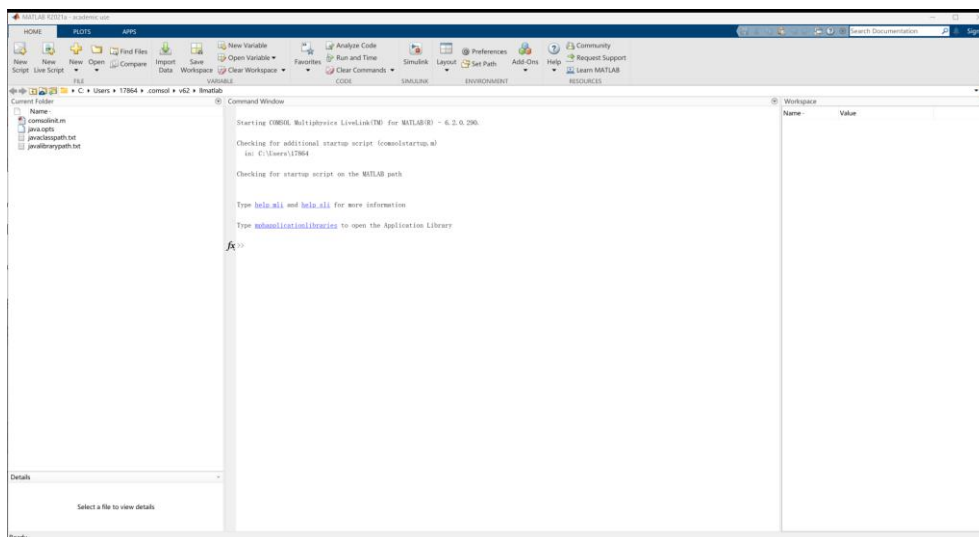
If COMSOL Multiphysics 6.2 with LiveLink for MATLAB is installed, follow the steps below to launch the MATLAB environment:

Step 1: Double-click the desktop icon of "COMSOL Multiphysics 6.2 with MATLAB" to open the command line terminal. A username and password are required for the first launch, and no re-verification is needed for subsequent use.



Step 2: COMSOL will automatically launch and load the MATLAB environment. Wait until the MATLAB main interface (including the command window, workspace and toolbar) is fully loaded.

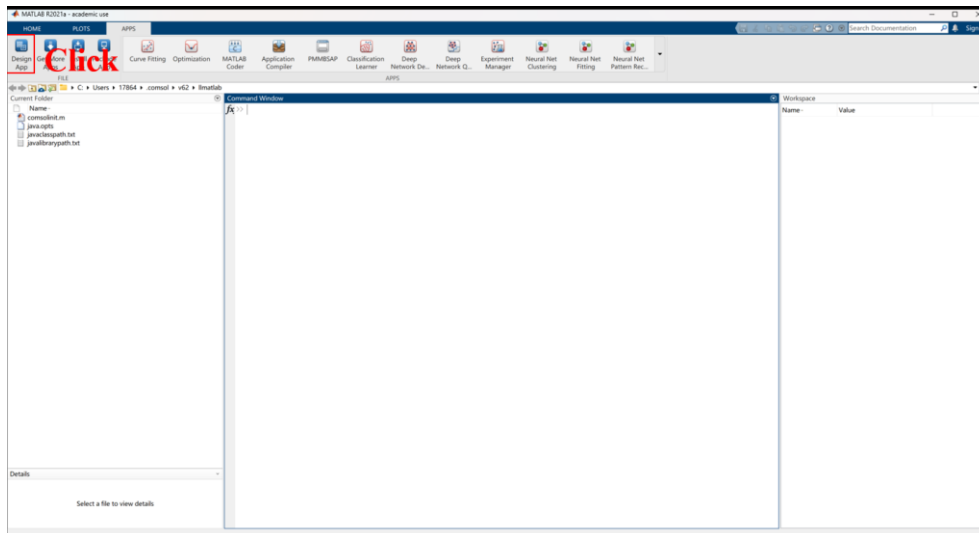
Step 3: Confirm that the initialization success information of COMSOL LiveLink is displayed in the MATLAB command window.



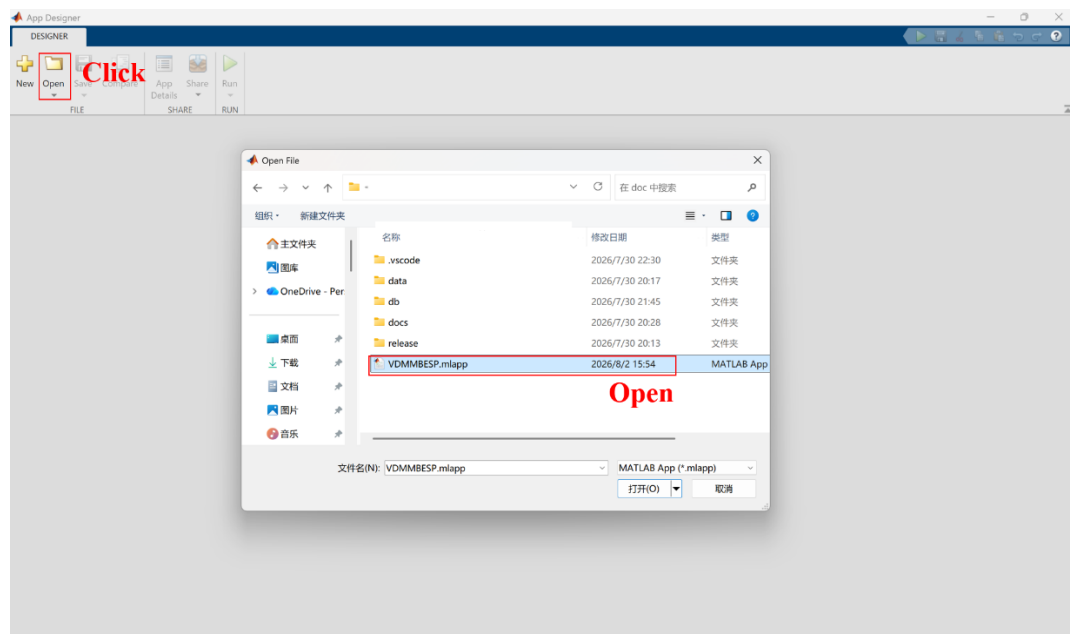
2.3 Software Launch

2.3.1 Launch via Source Code (Recommended)

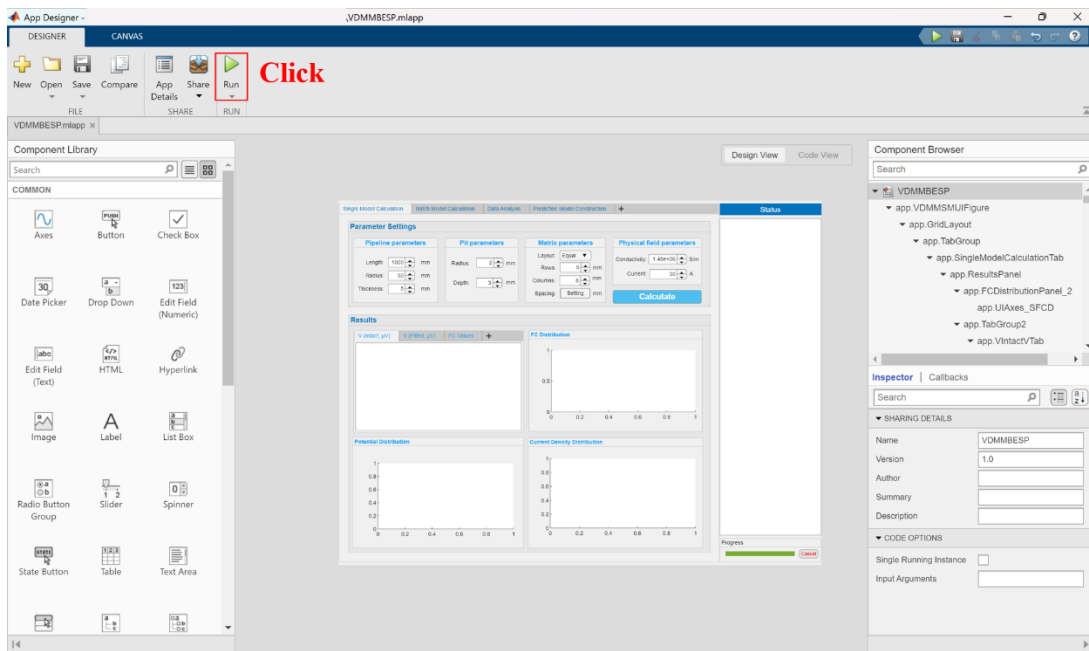
Step 1: Open MATLAB App Designer.



Step 2: Import and open the **VDDMMBESP.mlapp** file.

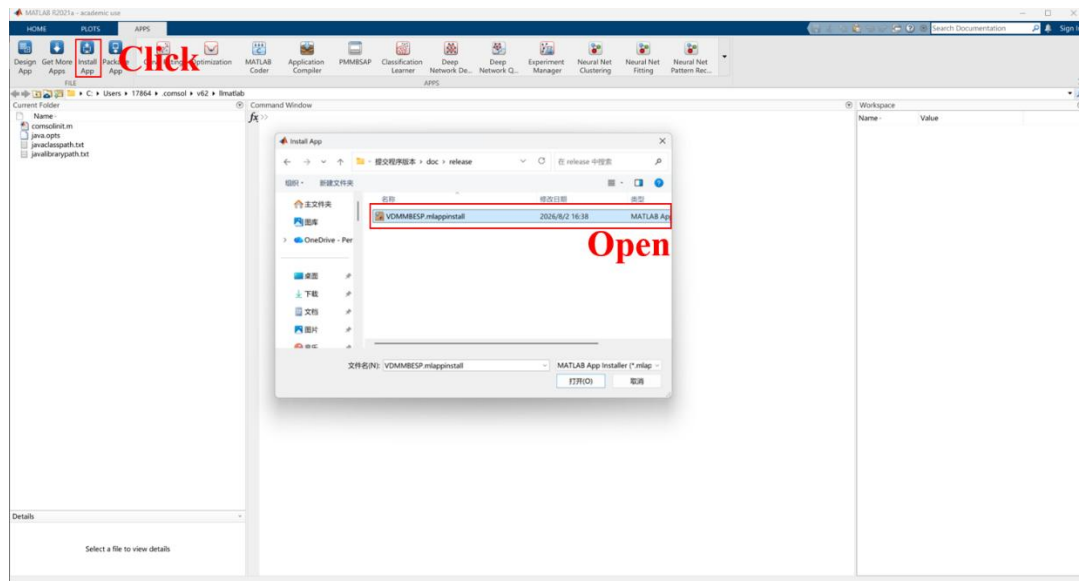


Step 3: Click the **Run** button to start the software.

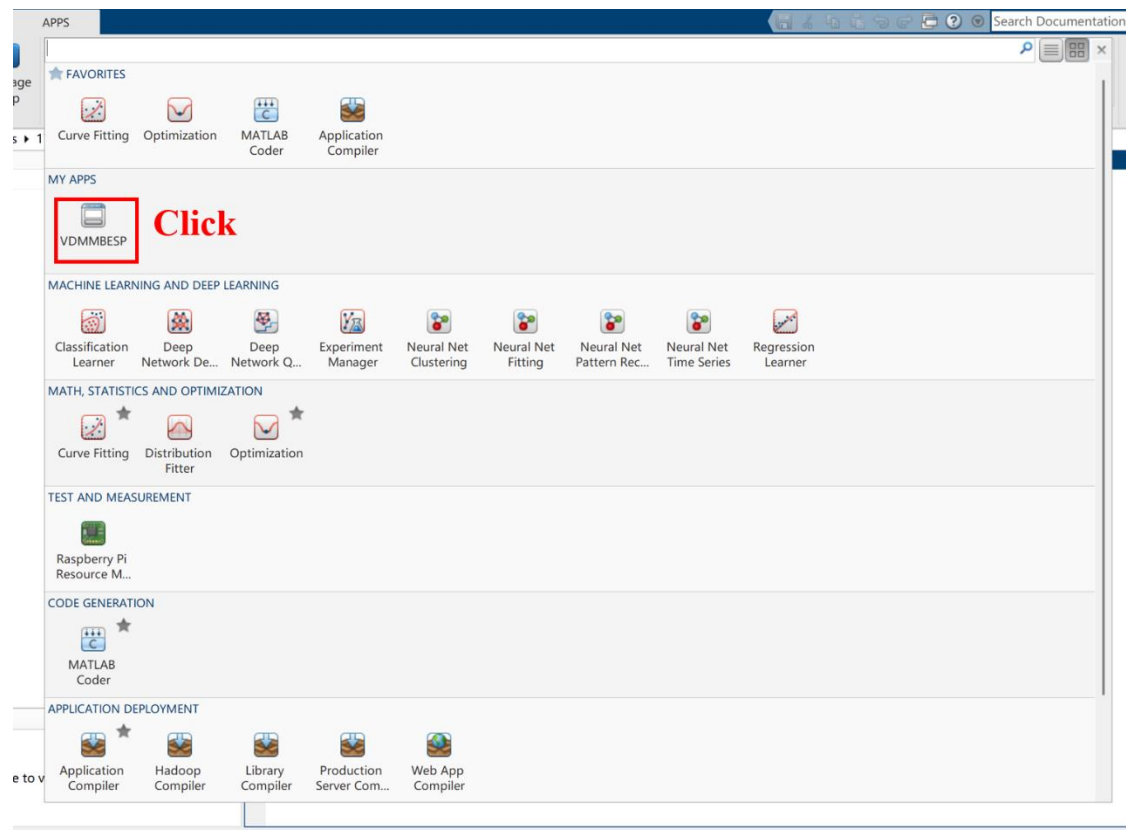


2.3.2 Launch via Packaged App

Step 1: Click **Install App** and import the "release/VDMMBESAP.mlappinstall" file.



Step 2: After installation, locate the VDMMBESAP icon in the **APPS** list and click it to run the application.



2.4 Database Configuration (Optional)

Configure the database if you need to store batch calculation results in MySQL for long-term storage and multi-user shared querying. This section can be skipped if only Excel and MAT file export functions are required.

2.4.1 Create Database and Tables

Open the Windows Terminal, connect to the MySQL database, and create the corresponding database and data tables. The SQL file for table creation is provided at `..\db\simulation_records.sql`.

```
C:\WINDOWS\system32\cmd. x + -
Microsoft Windows [版本 10.0.26200.8875]
(c) Microsoft Corporation. 保留所有权利。

C:\Users\17864>mysql -u root -p 1. Login to the database
Enter password: *****
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 47
Server version: 8.0.26 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE IF NOT EXISTS pipepitssim CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci; 2. Create a database
Query OK, 1 row affected (0.00 sec)

mysql> use pipepitssim; 3. Using the database
Database changed
mysql> source oc\db\simulation_records.sql 4. Create table structure
Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

2.4.2 Configure ODBC Data Source

Open the Windows Terminal, execute the command

C:\Windows\SysWOW64\odbcad32.exe to launch the ODBC Data Source

Administrator, and add a new data source following the graphical instructions.



3. Usage Guide

The graphical interface consists of four tabs, corresponding to different stages of the simulation and analysis workflow:

Single Model Calculation: Perform dual-model simulation with a complete set of fixed parameters to view the voltage matrix, *FC* distribution, electric potential, and current density field.

Batch Model Calculation: Automatically scan multiple groups of parameters to generate large-scale *FC* samples and export data.

Data Analysis: Load data from multiple sources, filter data by parameters, visualize single-model *FC* results, and conduct correlation analysis and regression fitting.

Prediction Model Construction: Train machine learning models including BP, RF, GBDT, and MVR to realize pitting parameter prediction based on *FC* values and output model evaluation metrics.

3.1 Single Model Calculation

This module is used to conduct a complete electric field simulation calculation for a set of fixed parameter values.

3.1.1 Parameter Definitions

The definitions of relevant simulation parameters are shown in the table below:

Type	Parameter Name	Abbreviation	Unit	Description
Pipeline parameters	Length	PipL	mm	Pipeline length
	Radius	PipR	mm	Pipeline radius
	Thickness	PipT	mm	Pipeline wall thickness
Pit parameters	Radius	Pitr	mm	Pitting radius
	Depth	Pith	mm	Pitting depth
Matrix parameters	Rows	Rows	/	Row count of electrode matrix
	Columns	Columns	/	Column count of electrode matrix
	Spacing	Spacing	mm	Electrode matrix spacing
Physical field parameters	Conductivity	/	S/m	Electrical conductivity of material
	Current	/	A	Excitation current value

3.1.2 Electrode Matrix Configuration

VDMMBESP supports two electrode deployment schemes, which can be switched via the **Layout** dropdown box:

Equal: After setting the Rows, Columns and Spacing parameters, the software automatically calculates the three-dimensional coordinates of each electrode on the pipeline surface.

Customize: Click the **Settings** button to pop up the **Electrode coordinate Table** dialog. Users can manually specify the 3D coordinates (x, y, z) of each electrode or import coordinate data from CSV files.

The image displays three 'Matrix parameters' dialog boxes and one 'Electrode Coordinate Table' dialog box.

The first 'Matrix parameters' dialog box shows 'Layout' set to 'Equal', 'Rows' set to 'Equal', 'Columns' set to 'Customize', and 'Spacing' set to 30 mm.

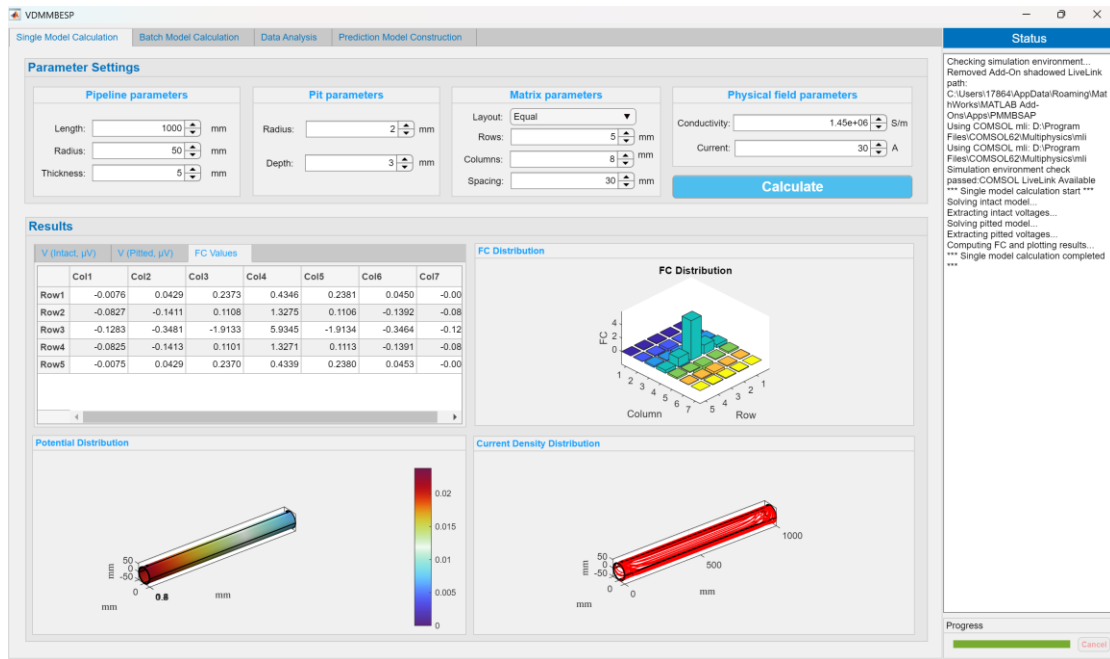
The second 'Matrix parameters' dialog box shows 'Layout' set to 'Equal', 'Rows' set to 5 mm, 'Columns' set to 8 mm, and 'Spacing' set to 30 mm.

The third 'Matrix parameters' dialog box shows 'Layout' set to 'Customize', 'Rows' set to 5 mm, 'Columns' set to 8 mm, and a 'Setting' button.

The 'Electrode Coordinate Table' dialog box is a table with 10 columns (Column 1 to Column 10) and 4 rows (x, y, z, and an empty row). The table is currently empty, with all cells containing 0. Below the table are three buttons: 'Load CSV File', 'Export to CSV', and 'Save to App Variable'.

3.1.3 Running Simulation

Click the **Calculate** button to execute the electric field distribution simulation under the currently set parameters. The output results include the voltage values of non-corroded and pitting-corroded pipelines, the FC values and distribution of pitting-corroded pipelines, as well as the cloud maps of pipeline potential and current density distribution.



3.2 Batch Model Calculation

The batch calculation module supports automatic cyclic simulation within the parameter space to generate massive sample data efficiently.

3.2.1 Export Options Configuration

Configure export parameters and file save paths in the **Export Options** and **File Save Path** panels before starting batch calculation:

Checkbox Option	Description	Remark
Export to Excel	Save parameters and FC results as .xlsx table files	Requires a customized save path and file name
Export to MAT-file	Save complete calculation results as .mat files	Require customized save path and file name
Save Report	Generate text reports with titles, timestamps and operation logs	Require customized save path and report name
Save to Database	Write simulation data to MySQL simulation_records table	Valid only after successful database connection
Select All	Enable all export functions with one click	/

3.2.2 Calculation Modes

Full Scan: Perform Cartesian product traversal and simulation for all parameter values.

Combination: Conduct calculation based on one-to-one matching of pipeline radius and wall thickness. When the **Enable Scan** checkbox is checked, full traversal calculation of pitting parameters will be additionally performed.

3.2.3 Batch Parameter Input Format

The batch parameter input box supports multiple input formats for multi-group parameter values, as shown in the table below:

Input Type	Input Content	Calculated Parameter Values
Space-separated	1 2 3 4	[1, 2, 3, 4]
Comma-separated	1, 2, 3, 4	[1, 2, 3, 4]
Colon expression	1:1:4	[1, 2, 3, 4]
Mixed format	1, 2:1:4	[1, 2, 3, 4]

3.2.4 Running Batch Calculation

Click the **Calculate** button to start batch calculation. The real-time progress percentage, elapsed time and operation logs can be viewed during the calculation process. After the batch calculation is completed, the **Results** panel provides two levels of data viewing functions:

Summary View: Display brief information of all calculation records.

Detailed View: Show complete and detailed data of batch calculation results.

The screenshot displays the VDMMBESP software interface. The top menu bar includes 'Single Model Calculation', 'Batch Model Calculation', 'Data Analysis', and 'Prediction Model Construction'. The 'Batch Model Calculation' tab is active.

Result Saving section:

- Export Options:** Checkboxes for 'Select All', 'Export to Excel', 'Export to MAT-file', 'Save Report', and 'Save to Database' are all checked.
- File Save Path:** Path is 'E:\data', File Name is 'data', and Report Name is 'report'.
- Database Configuration:** Data Source is 'test01', Username is 'root', Password is '123456'. A green 'Connect' button is visible.

Batch Calculation Parameters section:

- Calculation Mode:** Radio buttons for 'Full Scan' (selected) and 'Combination'.
- Pipeline parameters:** Length: 1200 mm, Radius: 50.10:100 mm, Thickness: 5.0:5.7 mm.
- Pits parameters:** Radius: 2.0:5.4 mm, Depth: 2.0:5.4 mm. An 'Enable Scan' checkbox is present.
- Matrix parameters:** Layout: Equal, Rows: 3, Columns: 4, Spacing: 15.5:40 mm.

Calculate button is located below the Pipeline parameters.

Results section:

Summary table:

Total Models	Calculation Time	Max Voltage (Intact)	Max Voltage (Pit)	Max FC	Min FC
4500	7 h 46 min	554.5986	564.5286	119.5668	-38.4977

Detailed results table (Index 1 to 6):

Index	PipeL	PipeR	PipeS	pitr	pith	rows	cols	spacing	sigma	current	FC_1	FC_2	FC_3	FC_4
1	1200	50	5	2	2	3	4	15	1450000	30	0.1811	3.1635	0.1801	-4.4286
2	1200	50	5	2	2	3	4	20	1450000	30	0.1149	1.7933	0.1135	-2.6268
3	1200	50	5	2	2	3	4	25	1450000	30	0.0833	1.1594	0.0828	-1.6987
4	1200	50	5	2	2	3	4	30	1450000	30	0.0670	0.8138	0.0667	-1.1755
5	1200	50	5	2	2	3	4	35	1450000	30	0.0570	0.6058	0.0567	-0.8554
6	1200	50	5	2	2	3	4	40	1450000	30	0.0504	0.4711	0.0501	-0.6493

On the right side, a **Status** panel shows the progress of the calculation, with a progress bar and a 'Cancel' button at the bottom.

3.3 Data Analysis

The data analysis tab provides a complete statistical analysis workflow, including data loading and screening, correlation analysis, univariate regression and multivariate regression modeling.

3.3.1 Data Loading and Selection

Select the data source type via the **Source** dropdown box in the **Data Selection** panel:

Type	Description
Batch	Read results of batch calculation in the current session
Excel	Import data from .xlsx files
MAT	Import data from .mat files
Database	Query MySQL data by parameter range (database connection required)

Click the **Load** button to pop up the data filtering dialog. Users can screen batch calculation records according to custom conditions, and select to display all data or specified row ranges via the **Data Range** dropdown box. Click **Save** to confirm and store the filtered data. **Each analysis method requires independent data selection and variable control: multiple values are assigned only to the variable of interest, and all other parameters are set to a single fixed value.**

Select Data (Excel)

Pipeline (min ~ max)

Length (L): 1200 ~ 1200

Radius (R): 50 ~ 90

Thickness (S): 5 ~ 7

Pitting (min ~ max)

Radius (r): 2 ~ 4

Depth (h): 2 ~ 4

Electrode

Rows: 3 ~ Cols: 4

Spacing: 15 ~ 40

Physical Field (single value)

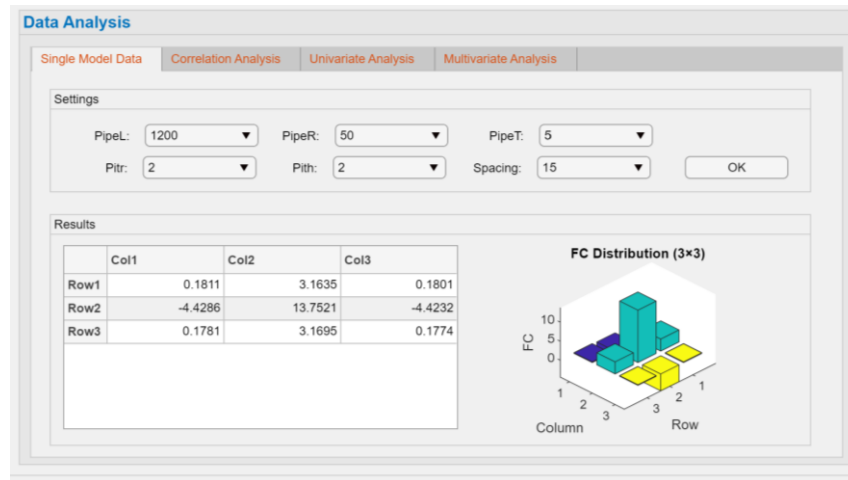
Sigma: 1450000 ~ Current: 30

Confirm Cancel

3.3.2 Selection of Data Analysis Methods

(1) Single Variable Data View

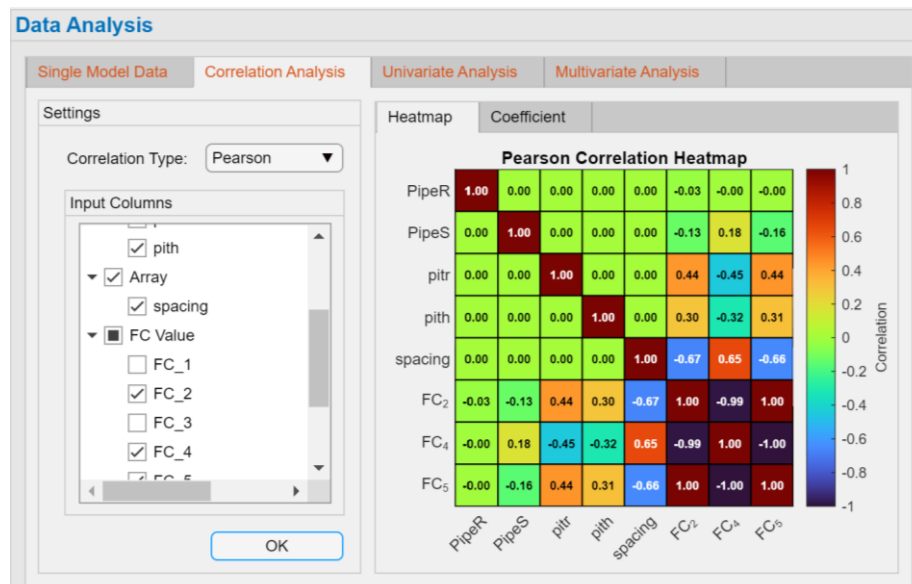
Select the target parameter combination via the dropdown box and click **OK**. The software will display the FC matrix in a table and visualize its distribution with a 3D column chart.



(2) Correlation Analysis

The correlation analysis module quantifies the statistical correlation between input parameters and FC output values, identifying the dominant influencing parameters. Operation steps are as follows:

- Check the target variables in the **Input Columns** selection tree
- Select a correlation algorithm (Pearson or Spearman) via the **Correlation Type** dropdown box
- Click **OK** to execute analysis
- Output results include a correlation matrix heatmap and a correlation coefficient table

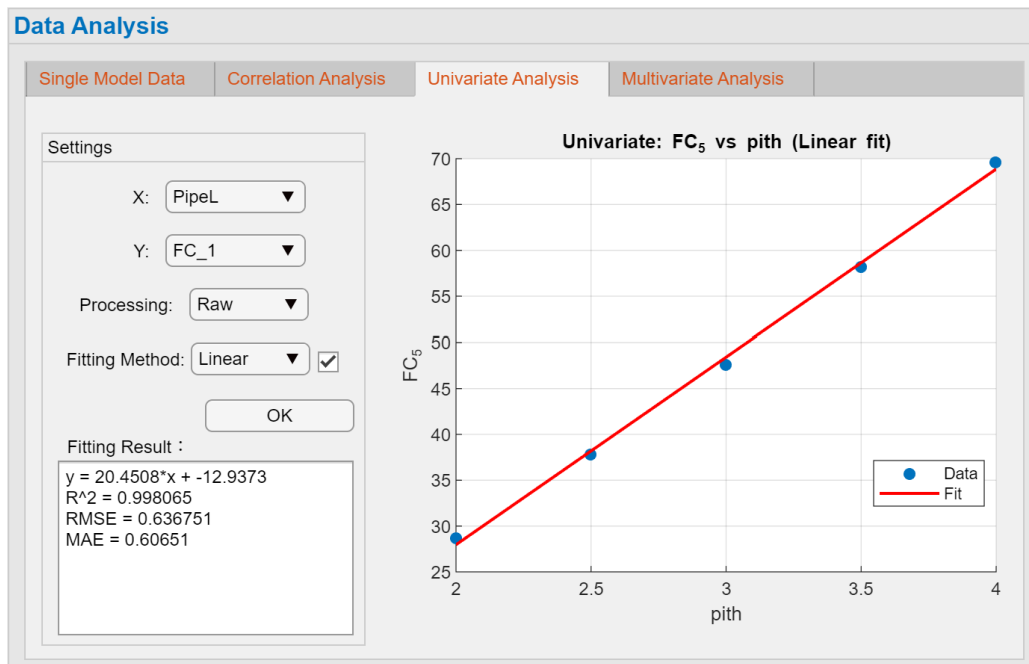


(3) Univariate Analysis

The univariate analysis module explores the quantitative relationship between a single input parameter and a single FC output value. The configurable parameters are as follows:

- **X:** Independent variable (single pipeline/pitting parameter)
- **Y:** Dependent variable (*FC* value)
- **Processing:** Data preprocessing modes (Raw / Logarithm / Reciprocal)
- **Enable Fitting:** Check to activate curve fitting function
- **Fitting Method:** Optional Linear or Polynomial fitting

The output includes a scatter plot (overlaid with red fitting curves after fitting activation), and quantitative evaluation results including fitting equation, R^2 , RMSE and MAE.

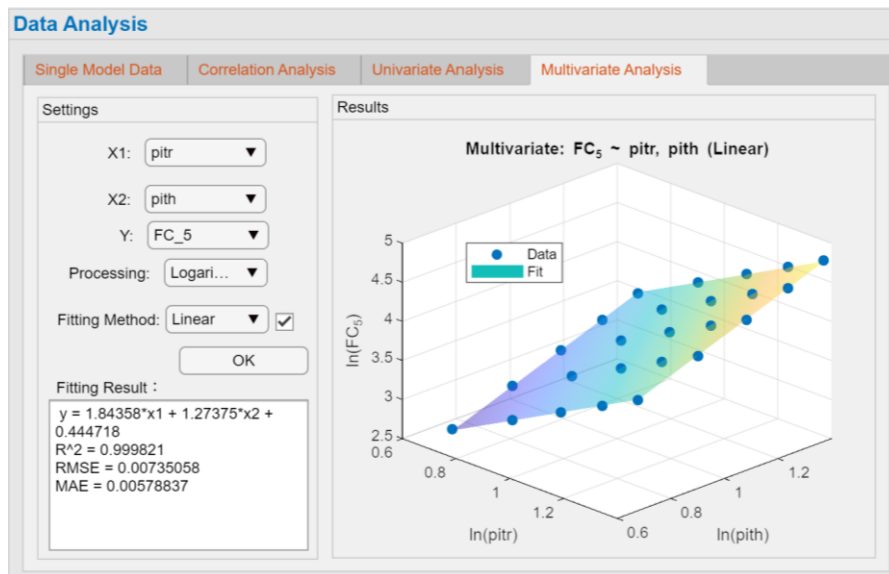


(4) Multivariate Analysis

As an extension of univariate analysis, this module supports joint regression analysis of two independent variables on one dependent variable. The configurable parameters are as follows:

- **X1, X2:** Two different independent variables
- **Y:** Dependent variable (FC detection value)
- **Processing:** Consistent with univariate analysis, acting on all variables
- **Enable Fitting:** Check to activate surface fitting function
- **Fitting Method:** Consistent with univariate analysis

The output includes a 3D scatter plot. A translucent fitting surface will be overlaid when fitting is enabled, together with the complete regression equation and evaluation metrics (R^2 , RMSE, MAE).



3.4 Prediction Model Construction

This module implements a data-driven, machine learning–based approach for inverting pitting corrosion parameters.

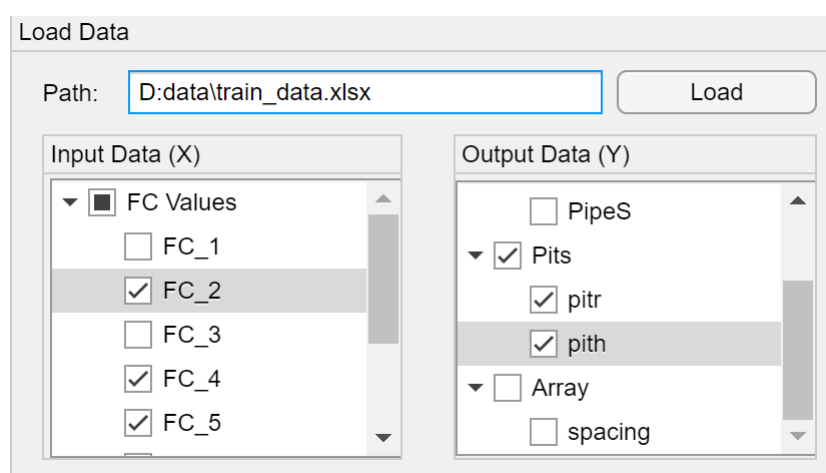
3.4.1 Data Preparation and Loading

Click **Load** to import Excel files with standard data column structures. The data import and filtering process is consistent with Section 3.2.1.

Select variables via the dual selection trees:

Input Data (X): Model input feature variables

Output Data (Y): Model prediction target variables



3.4.2 Model Selection and Hyperparameters

Select the required machine learning algorithm via the **Models** dropdown box, and adjust the corresponding hyperparameters in the **Settings** tab. Additionally, the interface provides noise

configuration options via two checkboxes to evaluate model robustness. The **N** checkbox adds noise exclusively to the **test set**. In contrast, the **NTrain** checkbox injects noise into **both the training set and the test set**.

Model Select

Models: BP TRatio: 0.8 SD: 0.03 ☐ N ☐ NTrain **Train**

Settings **Model Structure Flowchart**

epochs: 1000 lr: 0.01 goal: 0.0001

10

MVR (Multiple Linear Regression) configuration steps:

- Set the number of regression equations (N) via the numeric spinner;
- Select the dependent variable (Y) and two different independent variables (X1, X2) for each equation via the dropdown list;
- Select the data processing mode (Raw / Logarithm / Reciprocal); inverse transformation will be performed after coefficient estimation and prediction.

Model Select

Models: SER TRatio: 0.8 SD: 0.03 ☐ N ☐ NTrain **Train**

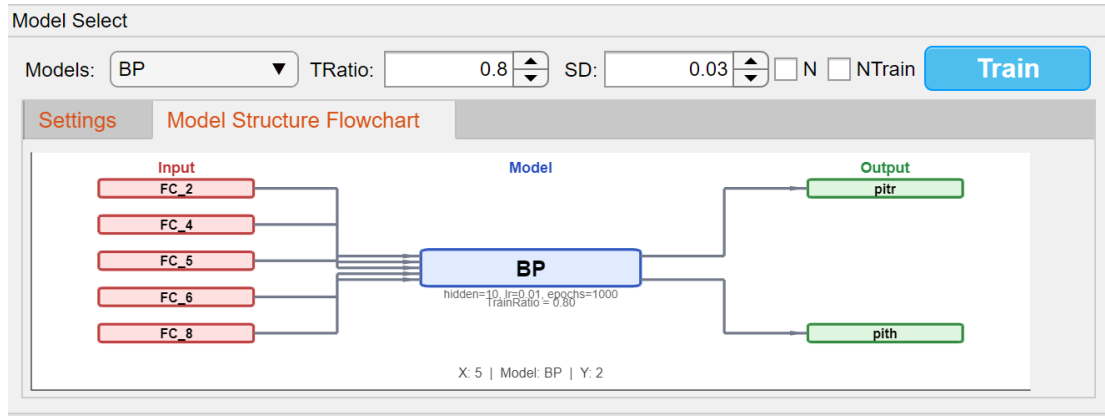
Settings **Model Structure Flowchart**

Processing: Raw Num: 2

Y	X1	X2
FC_2	pitr	pith
FC_5	pitr	pith

3.4.3 Training and Evaluation

Click the **Train** button to start model training, and the real-time model structure flowchart will be displayed on the interface.

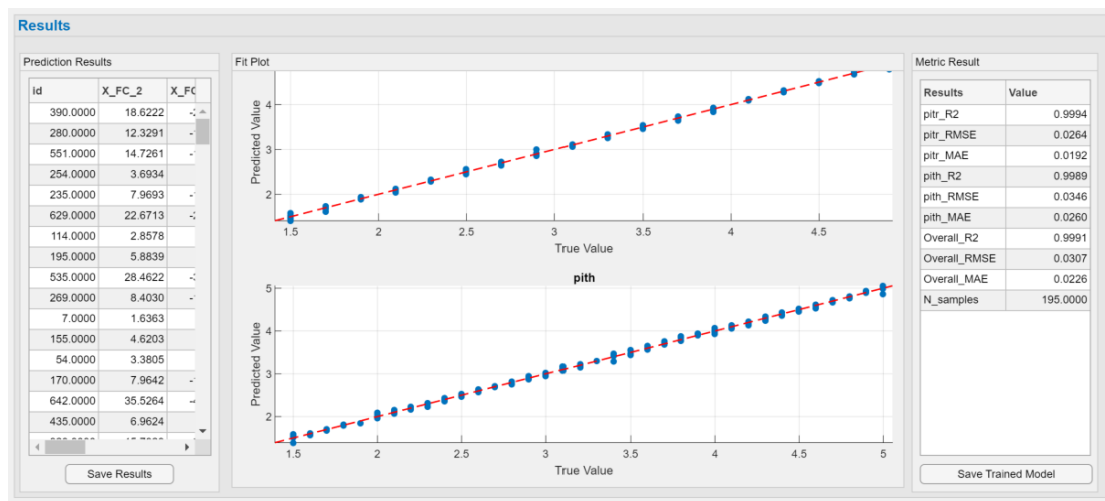


After training, the **Results** panel outputs the following contents:

- (1) **Prediction Results:** Input features, true values and predicted values of test samples
- (2) **Metric Results:** Quantitative evaluation metrics (R^2 , RMSE, MAE) of each output variable
- (3) **Fit Plots:** Scatter plots of true values vs. predicted values, with the red dashed line representing the $y=x$ reference line

Post-training operations:

- **Save Trained Model:** Save the trained PredTrainedModel structure as a .mat file;
- **Save Results:** Export prediction data and evaluation metrics as Excel files.



3.5 Model Prediction (Optional)

The saved prediction model can be adopted for the prediction of pitting parameters. Relevant executable programs are stored in the `./prediction` directory, and detailed operation instructions are provided in the `README.md` file within this folder.